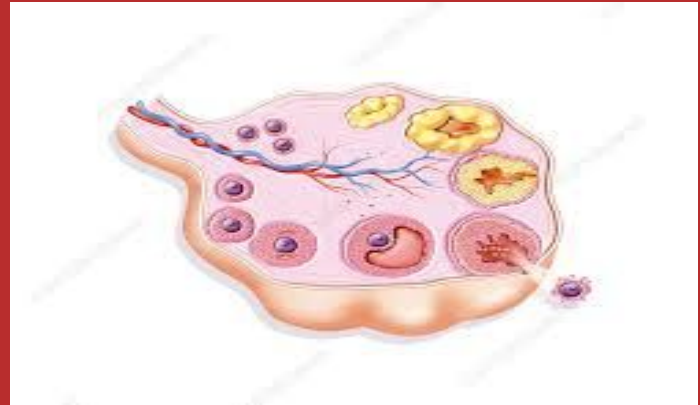
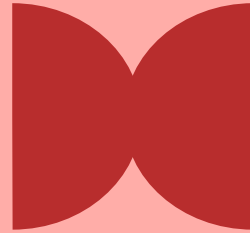




PCOS

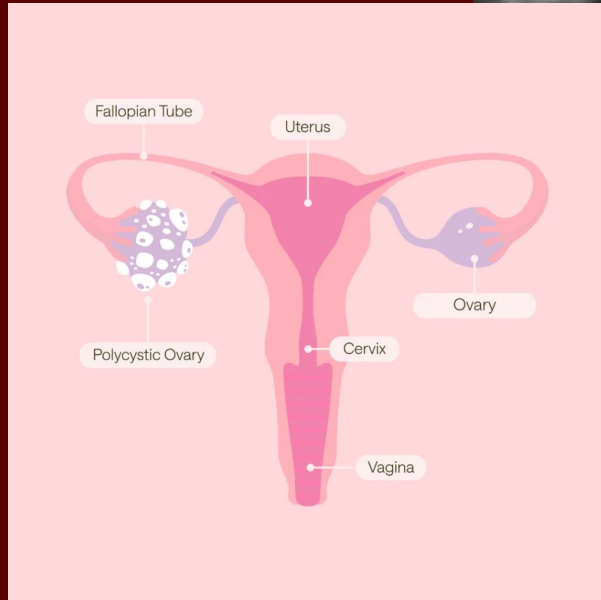
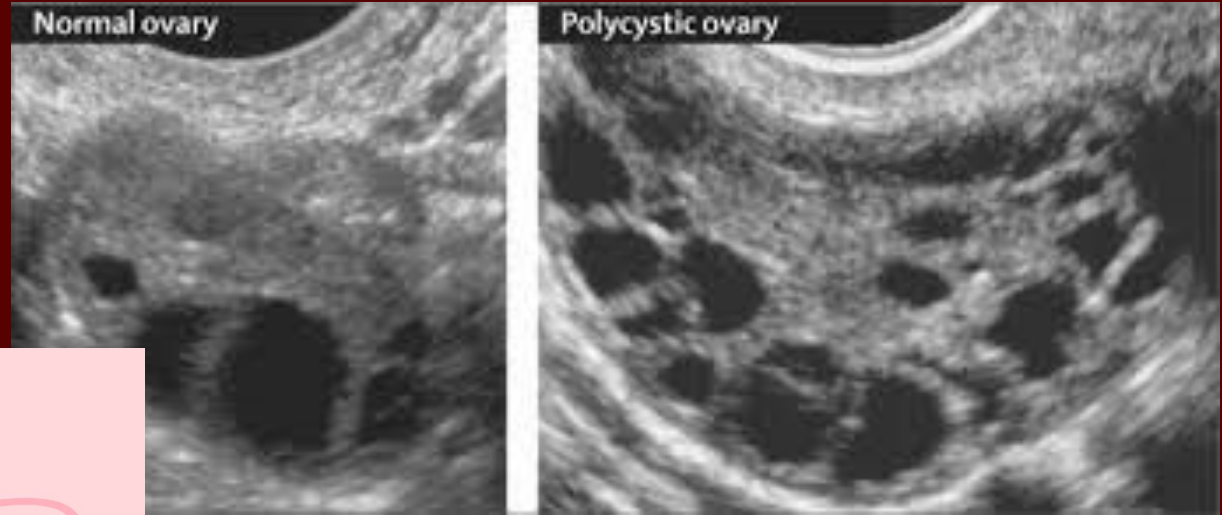
By: Samantha Bell



Background



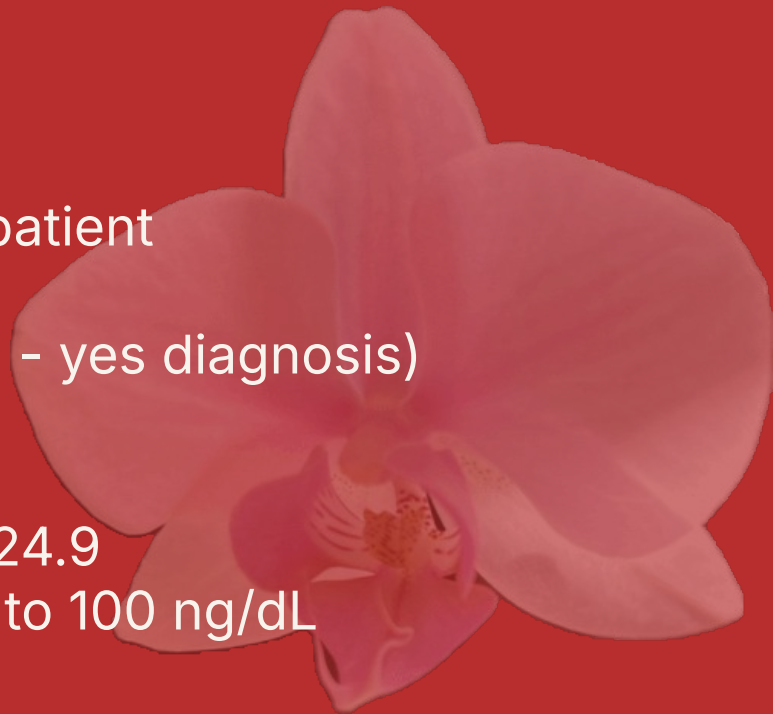
- Polycystic Ovarian Syndrome - meaning you got loads of cysts on your ovaries!
- Hormonal disorder/imbalance caused by either excess androgen production by the ovaries, insulin resistance, inflammation, or genetics
- Symptoms include irregular periods, excess hair, weight gain, fatigue, depression, infertility, and much more
- Very under-researched, only 13% of women are diagnosed, estimated that 70% of others are undiagnosed



Ovaries!

About the data

- 1000 entries, each from a different patient
- PCOS_diagnosis (0 - no diagnosis, 1 - yes diagnosis)
- Age (years), ranging from 18 to 45
- BMI, ranging from 18 to 35
 - Healthy BMI is considered 18.5-24.9
- Testosterone level, ranging from 20 to 100 ng/dL
 - normal level is 15-70 ng/dL
- Menstrual irregularity (0 - no, 1 - yes)
- Antral follicle count, ranging from 5 - 30
 - fluid-filled sac on the ovary containing a developing egg



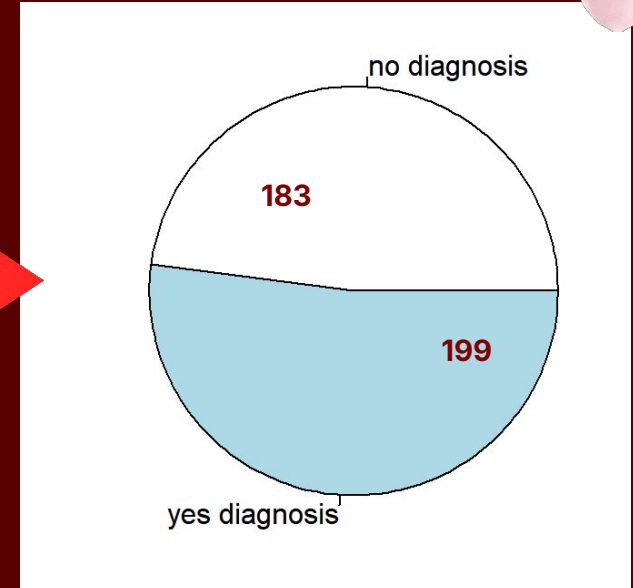
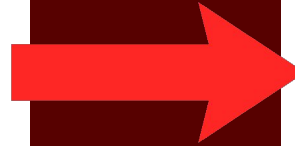
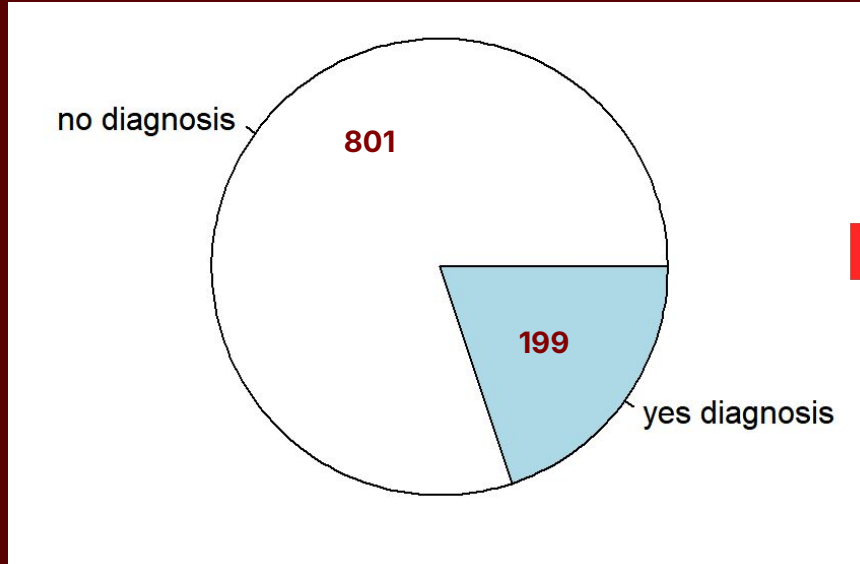
Objective:

To compare the accuracy of

- Logistic regression
- LDA
- Decision tree
- Random forest
- Bagging
- Naive bayes



Balancing the data



Total = 382

Choosing predictor variables



Logistic regression

```
Call:  
glm(formula = PCOS_Diagnosis ~ ., family = "binomial", data = pcos)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-65.55789	2656.89094	-0.025	0.980
Age	-0.01107	0.02293	-0.483	0.629
BMI	0.50589	0.06587	7.680	1.59e-14 ***
Menstrual_Irregularity	23.33562	1328.44429	0.018	0.986
Testosterone_Level.ng.dL.	0.05992	0.01078	5.558	2.73e-08 ***
Antral_Follicle_Count	0.14034	0.03506	4.003	6.26e-05 ***



Backward Step

Step: AIC=178.49

PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. +
Antral_Follicle_Count

	Df	Deviance	AIC
<none>		168.49	178.49
- Antral_Follicle_Count	1	187.41	195.41
- Testosterone_Level.ng.dL.	1	211.25	219.25
- BMI	1	290.90	298.90
- Menstrual_Irregularity	1	361.02	369.02



Forward Step

Step: AIC=178.49

PCOS_Diagnosis ~ Menstrual_Irregularity + BMI + Testosterone_Level.ng.dL. +
Antral_Follicle_Count

	Df	Deviance	AIC
<none>		168.49	178.49
+ Age	1	168.26	180.26

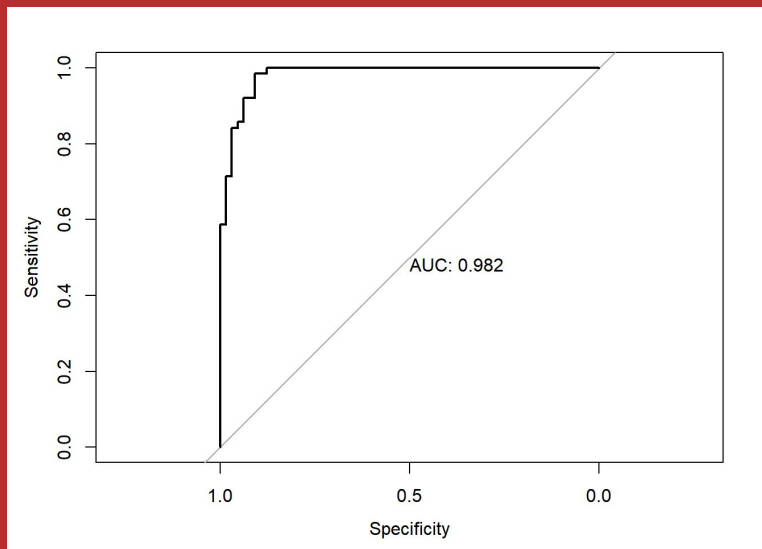


Begin testing models!



Split the data:
 $\frac{2}{3}$ training
 $\frac{1}{3}$ testing

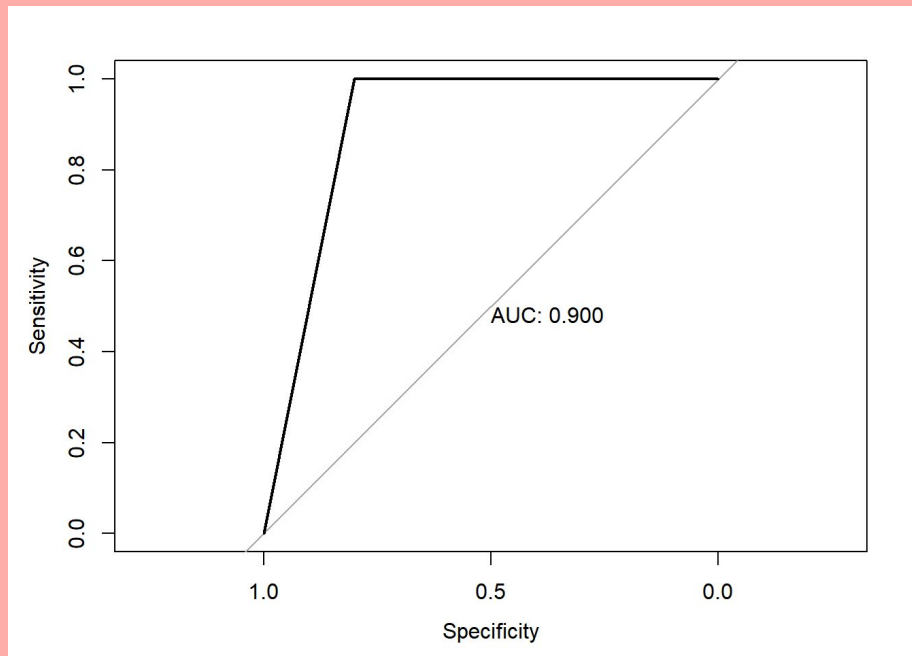
Logistic regression



p	0	1
0	59	3
1	6	60

accuracy : 0.9297

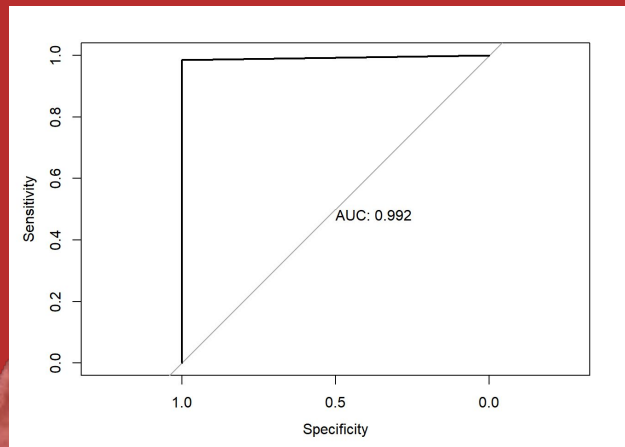
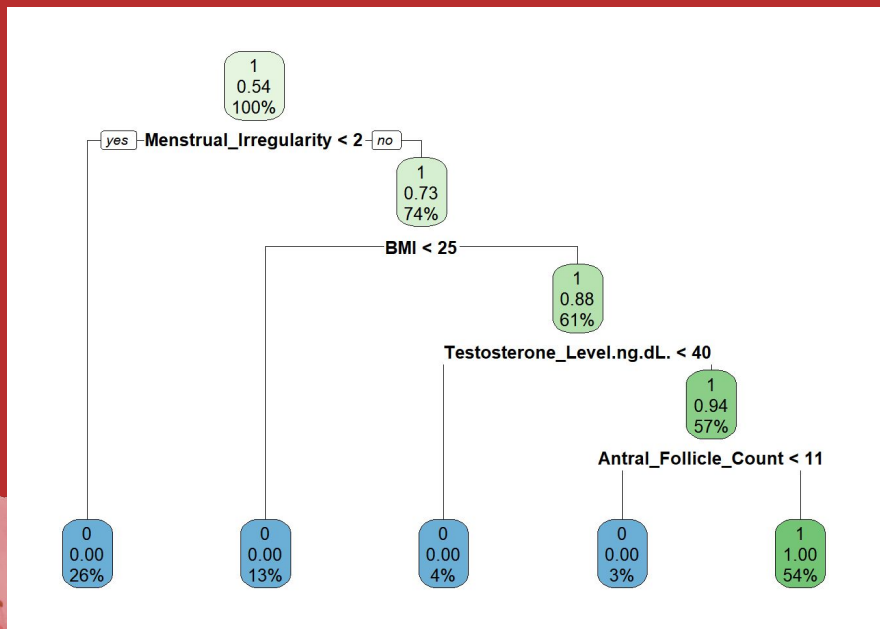
LDA



p	0	1
0	52	0
1	13	63

accuracy : 0.8984

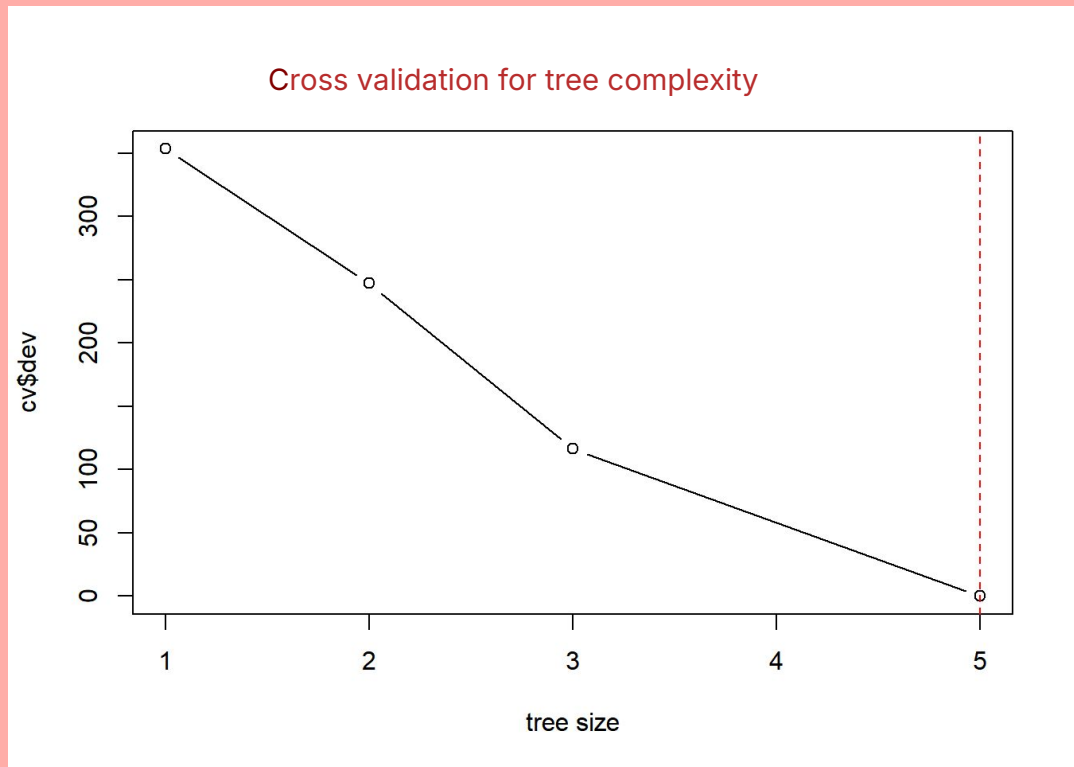
Decision tree



p	0	1
0	65	1
1	0	62

accuracy : 0.9923

Pruning

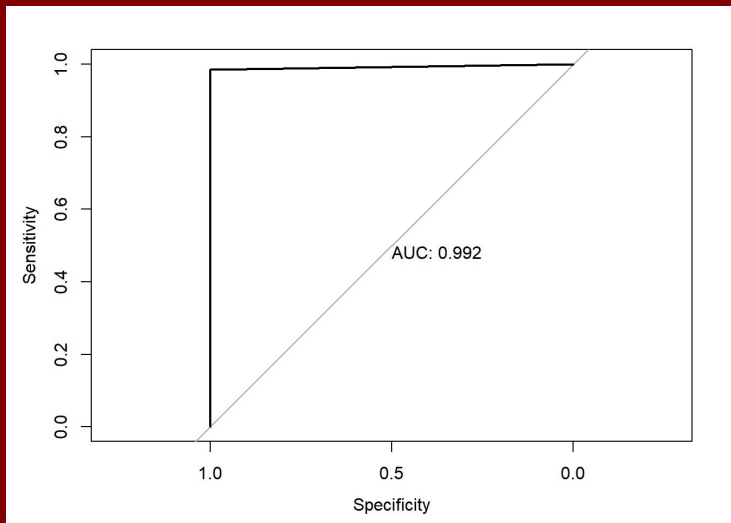


No need to change
my tree!



Random forest

```
rf = randomForest(PCOS_Diagnosis ~ BMI + Menstrual_Irregularity + Testosterone_Level.ng.dL. + Antral_Follicle_Count, data = pcos_train, mtry = 2)
```

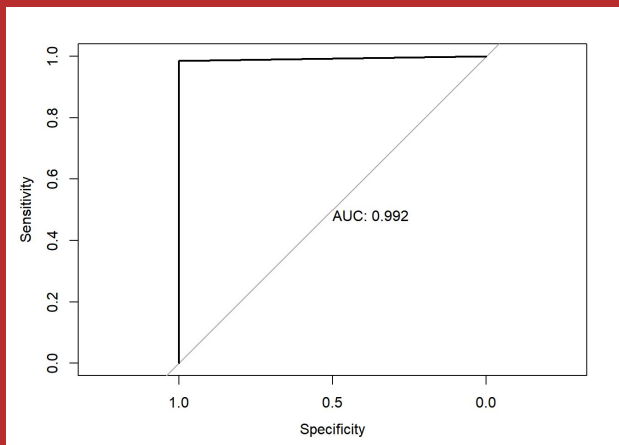


p	0	1
0	65	1
1	0	62

accuracy : 0.9923

Bagging

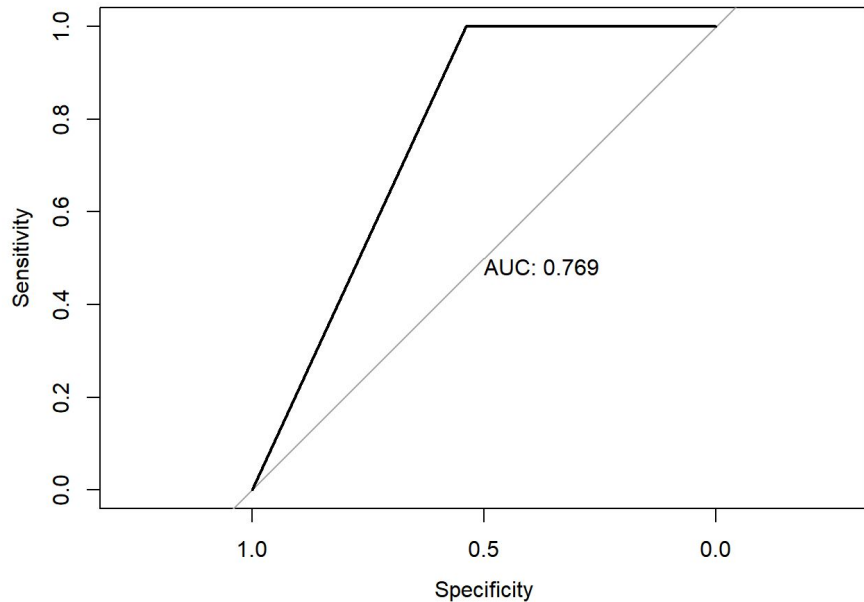
```
bag = randomForest(PCOS_Diagnosis ~ BMI + Menstrual_Irregula  
rity + Testosterone_Level.ng.dL. + Antral_Follicle_Count, da  
ta = pcos_train, mtry = 4)
```



p	0	1
0	65	1
1	0	62

accuracy : 0.9923

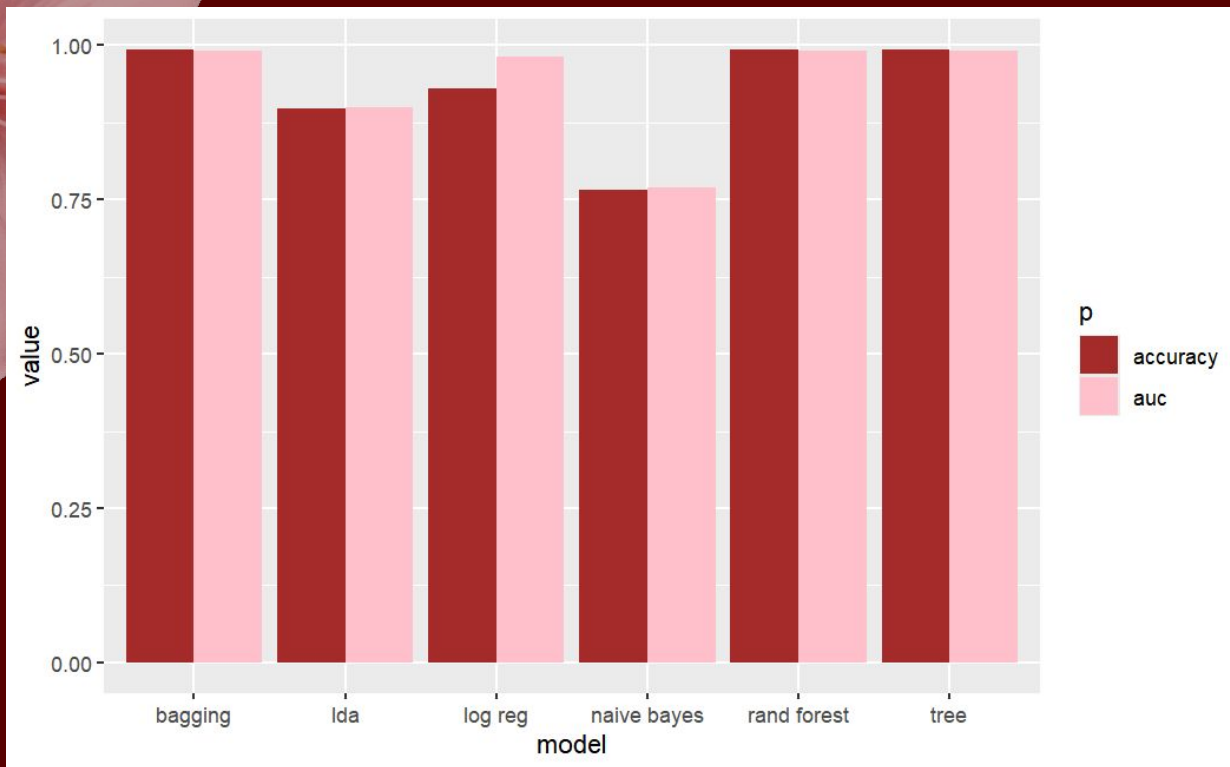
Naive bayes



p	0	1
0	35	0
1	30	63

accuracy : 0.7656

Comparison!



What to do next:

- Try to do QDA + other models in general
- Use a more complex data set with more predictors
- Measure AUC on full data set
- Test with interaction terms



Conclusion



Thank you!

